

Mobina Amrollahi

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Professional Summary

Ph.D. researcher with 3+ years building end-to-end systems for autonomous agents and a year owning fintech product delivery at a major Middle East e-commerce platform. Published in leading venues including ACM Transactions on Human-Robot Interaction, with 5 conference presentations. My work combines statistical, probabilistic, and data-driven methods to model human behavior from multimodal interaction data, yielding policies that let robots reason about people and act in socially acceptable ways.

Education

Ph.D. Computer Science, Iowa State University, Ames, IA Aug. 2023 – Aug. 2028
M.Sc. Industrial Engineering, Iowa State University, Ames, IA Aug. 2023 – Aug. 2026
B.S. Industrial Engineering, Sharif University of Technology, Iran 2018 – 2023

Technical Skills

Tools: Python (PyTorch, TensorFlow, scikit-learn, NumPy, pandas), R, C#, SQL, Unity, Git / GitHub

Modeling: Reinforcement Learning, Deep Learning (Behavioral Cloning), Unsupervised Learning (Clustering), Human Behavior Modeling, Motion Planning (Model-, Sampling-, & Learning-Based)

Experimental & Statistical Methods: Experimental Design (Full-Factorial), Human-Subject Study Design, IRB Submission & Handling, Validated Psychometric Instruments (NASA-TLX, Trust Scales), Statistical Hypothesis Testing, Mixed-Methods Analysis

Experience

Human-Robot Interaction Graduate Research Assistant, Iowa State University Aug. 2023 – Present

- Designed & implemented end-to-end simulation environments in Unity (C#), then ran three full-factorial experiments with 100 participants in total
- Trained a deep neural network (behavioral cloning) in Python to predict end-user actions & deployed the trained weights in Unity to tune the robot's policy in real time
- Developed a dynamics-agnostic integrating method for the network into the robot's decision-making
- Built reproducible cross-language analysis pipelines (Python, R) for multimodal time-series data
- Measured up to 24% faster human-robot task completion than the state of the art, with no collisions under the validated policy
- Presented findings at Toyota Technological Institute at Chicago & to industry partners such as John Deere

Junior to Associate Product Owner, Digikala, Iran Aug. 2021 – Sept. 2022

- Owned Bank Mellat's credit card integration on Digikala, a major Middle East e-commerce platform
- Defined delivery KPIs and coordinated execution across design, engineering, test, & release teams
- Performed exploratory analysis of customer data to inform product and business decisions

Selected Publications

M. Amrollahi, M. Dorneich, & J. Liu. [Measuring & Understanding Trust in Motion: Behavioral Archetypes in Human-Automated Guided Vehicle Interactions](#). *ACM Transactions on Human-Robot Interaction*, 2026.

A. Gomez-Hernandez*, **M. Amrollahi***, et al. [A Systematic Review of Human-in-the-Loop Adaptive Systems: Design, Implications, & Future Directions](#). *IEEE Transactions on Human-Machine Systems*, 2026.

M. Amrollahi, R. Wangira, & J. Liu. [Analysis & Modeling of Worker Trust in Automated Guided Vehicles for Manufacturing Workplace](#). *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 2025.

Selected Honors

Alphonse Chapanis Paper Award Runner-Up, Human Factors & Ergonomics Society 2025
NSF IISE Annual Conference Student Travel Award, NSF Award ID CMMI-2511912 (\$840) 2025
Harold and Shirley Reihman Graduate Scholarship, Iowa State University (\$3,000) 2023
UMN Provost Fellowship, University of Minnesota (\$3,300) 2023